Misspecification-Robust Inference for the Incremental Fit Indices CFI and TLI under an Estimated Weight

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Purpose

The companion notes fix the estimated-weight inference of a *single* quadratic-form criterion. `rmsea_asymptotics.tex` and the RMSEA path of `higher_order_discrepancy_misspec` treat the discrepancy $F = r'Wr$ itself; the CRMR path treats the correlation residual sum $G = r'V_0r$. Both reduce to one object: a statistic $T = N\hat{F}$, a bias-corrected noncentrality $\hat{\delta} = T - \operatorname{tr}(Q\Gamma_x)$ where Q is a profile Hessian and Γ_x the joint NACOV of the extended first-stage vector $x = (u, \gamma)$, and a sampling variance $\operatorname{Var}(T) = N\, g'\Gamma_x g$ obtained by the delta method from an envelope-score gradient g.

This note opens the program's *first incremental* (two-model) index. CFI and TLI are not functions of one criterion; they compare the fitted user model to the independence baseline,

$$\mathrm{CFI} = 1 - \frac{\max(0, T_u - df_u)}{\max(0, T_b - df_b)}, \qquad \mathrm{TLI} = \frac{T_b/df_b - T_u/df_u}{T_b/df_b - 1}. \tag{1}$$

Under an estimated weight neither T_u nor T_b is χ^2: each is a mixture $\sum_j \lambda_j \chi^2_1$ whose mean is the generalized degrees of freedom $\bar{Q} = \operatorname{tr}(Q\Gamma_x)$, not the nominal df. The single new statistical object is the *joint* law of (T_u, T_b). The headline is that both statistics are smooth functionals of the *same* x, so their joint covariance is one bilinear form in the *one* shared Γ_x:

$$\operatorname{Cov}(T_u, T_b) \;=\; N\, g_u'\, \Gamma_x\, g_b, \tag{2}$$

and CFI and TLI are then a ratio delta-method on the two correlated noncentralities. No new covariance machinery is needed beyond the second gradient

g_b and the second profile Hessian Q_b; the metric Γ_x is shared. Three consequences follow: an honest Wald interval for CFI (Corollary 1), a TLI interval that is the CFI interval rescaled by a generalized-df ratio (Corollary 2), and a baseline-dominated simplification (Remark) under which CFI's variance collapses to the RMSEA-side variance the program already computes, divided by the squared baseline noncentrality.

1 Setup

Write $x = (u, \gamma) \in \mathbb{R}^{2m}$ for the extended first-stage vector: the m polychoric sample moments u (thresholds and correlations, stacked in the `OrdinalStats` order) and the DWLS weight diagonal $\gamma = \operatorname{diag} \hat{\Gamma}$, $W = \operatorname{diag}(1/\gamma)$. Let

$$\sqrt{N}(\hat{x} - x_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_x),$$

with Γ_x the joint per-unit NACOV built from the stacked casewise influence $[\, g_i \mid \mathrm{IF}_i(\gamma) \,]$ exactly as in `ordinal_dwls_profile_rmsea`. Γ_x is a property of the data, not of any fitted model: *the user model and the baseline share it.*

For a model $M \in \{u, b\}$ (user, baseline) the minimum-discrepancy fit profiles its parameters out of

$$F_M(x) = d_M(x)' \, W \, d_M(x), \qquad d_M = \sigma_M(\hat{\theta}_M) - u,$$

giving statistic $T_M = N F_M$, profile Hessian Q_M (the $2m \times 2m$ extended object of `weighted_moment_profile_rmsea_estimated_weight`), generalized df $\bar{Q}_M = \operatorname{tr}(Q_M \Gamma_x)$, and bias-corrected noncentrality

$$\hat{\delta}_M \;=\; T_M - \bar{Q}_M. \tag{3}$$

The user model is the fitted CFA/SEM. The baseline is the independence model: free thresholds, all polychoric correlations fixed to zero (delta scaling fixes the variances). Its residual is $d_b = (\,0; -\operatorname{vech}_< \hat{R}\,)$ (thresholds profile to zero residual under diagonal W; the correlation block is the negative sample correlations), and T_b matches `ordinal_baseline_chi2`. Because σ_b is *linear* in the baseline thresholds, the baseline observed bread equals its Gauss–Newton bread and Q_b is exact.

Envelope gradient. Since $\hat{\theta}_M$ minimizes F_M, the $\hat{\theta}_M$-movement term in dF_M/dx vanishes (envelope theorem) and the gradient is the bare profile score, identical in form for both models:

$$g_M = \Big(\tfrac{\partial F_M}{\partial u},\ \tfrac{\partial F_M}{\partial \gamma}\Big) = \big(-2\,W d_M,\ -d_M^2/\gamma^2\big) \in \mathbb{R}^{2m}, \tag{4}$$

the γ block coming from $\partial W/\partial\gamma_k = -1/\gamma_k^2$ in $F_M = \sum_k d_{M,k}^2/\gamma_k$. Equation (4) is exactly the RMSEA envelope score, evaluated once at the user residual d_u and once at the baseline residual d_b.

2 Joint law of the two statistics

Theorem 1. *Under fixed misspecification of both models and the regularity of the single-criterion program, with $\hat{x} = \bar{x}$ the sample first-stage vector,*

$$\sqrt{N}\begin{pmatrix} F_u - F_{0,u} \\ F_b - F_{0,b} \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(0,\ \begin{pmatrix} g_u'\Gamma_x g_u & g_u'\Gamma_x g_b \\ g_b'\Gamma_x g_u & g_b'\Gamma_x g_b \end{pmatrix}\right).$$

Equivalently, with $V_{uu} = N g_u'\Gamma_x g_u$, $V_{bb} = N g_b'\Gamma_x g_b$, $V_{ub} = N g_u'\Gamma_x g_b$, the statistic pair satisfies $\mathrm{Var}(T_u) = V_{uu}$, $\mathrm{Var}(T_b) = V_{bb}$, $\mathrm{Cov}(T_u, T_b) = V_{ub}$, *which is* (2).

Proof. Each $F_M(\hat{x}) = F_{0,M} + g_M'(\hat{x} - x_0) + o_p(N^{-1/2})$ by a first-order expansion, with g_M the envelope gradient (4) (the profiled parameters contribute no first-order term). Stacking the two expansions and applying the single CLT $\sqrt{N}(\hat{x} - x_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_x)$ to the common $\hat{x}$ gives the bivariate limit; the off-diagonal is $g_u'\Gamma_x g_b$ because both rows read the same increment $\hat{x} - x_0$. Scaling by N ($T_M = N F_M$) gives the stated variances. $\square$

The mean corrections $\bar{Q}_M$ are $O_p(1)$ and, as in the single-criterion program, are treated as fixed in the variance (their sampling fluctuation is higher order). Hence $\mathrm{Var}(\hat{\delta}_u) = V_{uu}$, $\mathrm{Var}(\hat{\delta}_b) = V_{bb}$, $\mathrm{Cov}(\hat{\delta}_u, \hat{\delta}_b) = V_{ub}$.

3 CFI

In the regime where both models misfit, $\hat{\delta}_u > 0$ and $\hat{\delta}_b > 0$, the truncations in (1) are inactive and, writing the generalized-df noncentralities for df,

$$\widehat{\mathrm{CFI}} \;=\; 1 - r, \qquad r := \frac{\hat{\delta}_u}{\hat{\delta}_b}. \tag{5}$$

The population target is $\mathrm{CFI}_0 = 1 - \delta_{0,u}/\delta_{0,b} = 1 - F_{0,u}/F_{0,b}$ (the N cancels in the ratio).

Corollary 1. *The ratio gradient is* $\nabla r = \hat{\delta}_b^{-1}(1, -r)$*, so by the delta method*

$$\widehat{\mathrm{Var}}(\widehat{\mathrm{CFI}}) = \widehat{\mathrm{Var}}(r) = \frac{1}{\hat{\delta}_b^2}\Big(V_{uu} - 2\,r\,V_{ub} + r^2 V_{bb}\Big), \tag{6}$$

and a two-sided $1-\alpha$ *interval is* $\widehat{\mathrm{CFI}} \pm z_{1-\alpha/2}\,\widehat{\mathrm{Var}}(\widehat{\mathrm{CFI}})^{1/2}$*, clipped to* $[0, 1]$*.*

4 TLI

TLI is the same comparison taken per generalized degree of freedom. With $E[T_M] \approx \bar{Q}_M + \delta_M$, the misspecification-robust analogue of T/df is $T_M/\bar{Q}_M \approx 1 + \delta_M/\bar{Q}_M$, and substituting into the TLI form of (1) gives

$$\widehat{\mathrm{TLI}} = \frac{T_b/\bar{Q}_b - T_u/\bar{Q}_u}{T_b/\bar{Q}_b - 1} = 1 - c\,r, \qquad c := \frac{\bar{Q}_b}{\bar{Q}_u}. \tag{7}$$

Thus TLI is CFI's ratio r rescaled by the generalized-df ratio c; $\widehat{\mathrm{TLI}} = 1 - (1 - \widehat{\mathrm{CFI}})\,\bar{Q}_b/\bar{Q}_u$. Using $\bar{Q}$ rather than the nominal df keeps TLI internally consistent with the mixture mean that defines $\hat{\delta}$; when the weight is correctly specified $\bar{Q}_M \to df_M$ and (7) recovers the lavaan point value.

Corollary 2. *Treating* c *as fixed (the* $\bar{Q}_M$ *are deterministic mean corrections),*

$$\widehat{\mathrm{TLI}} = 1 - c\,r \implies \widehat{\mathrm{Var}}(\widehat{\mathrm{TLI}}) = c^2\,\widehat{\mathrm{Var}}(\widehat{\mathrm{CFI}}). \tag{8}$$

The TLI interval is the CFI interval scaled by $c = \bar{Q}_b/\bar{Q}_u$*: same* r*, same* $\widehat{\mathrm{Var}}(r)$*, one extra factor. (TLI is not clipped to* $[0, 1]$*; it can exceed* 1 *for* $\hat{\delta}_u < 0$*.)*

5 The baseline-dominated regime

Under any appreciable misspecification the independence model fits grossly, so $\hat{\delta}_b \gg \hat{\delta}_u$, $r \to 0$, and (6) collapses to its leading term:

$$\widehat{\mathrm{Var}}(\widehat{\mathrm{CFI}}) = \frac{V_{uu}}{\hat{\delta}_b^2}\Big(1 - 2r\,\tfrac{V_{ub}}{V_{uu}} + r^2\,\tfrac{V_{bb}}{V_{uu}}\Big) \approx \frac{V_{uu}}{\hat{\delta}_b^2} = \frac{\mathrm{Var}(T_u)}{\hat{\delta}_b^2}. \tag{9}$$

That is, to leading order CFI's sampling variance is the *user-model* statistic variance $V_{uu} = N g_u' \Gamma_x g_u$ already returned as `grad_var` by the RMSEA path, divided by the squared baseline noncentrality $\hat{\delta}_b^2$; the baseline acts as a nearly constant scale. The cross-term and baseline-variance contributions are $O(r)$

and $O(r^2)$ corrections. The full bivariate form (6) is retained in code for honesty, and it must be: the Monte-Carlo check (`tests/checks/ordinal_cfi_inference`) finds the ratio $(V_{uu}/\hat{\delta}_b^2)/\widehat{\mathrm{Var}}(\widehat{\mathrm{CFI}})$ falling from 0.99 at weak misfit to 0.35 at strong misfit, i.e. (9) is accurate only near correct specification, where "CFI inference is a rescaling of RMSEA inference" holds; as misfit grows the cross-term and baseline-variance corrections take over. The simplification is intuition, not the computation.

6 Estimated versus fixed weight

Zeroing the γ blocks of Γ_x (drop the IF(γ) rows/columns) gives the fixed-weight comparator: the g_M γ-channel in (4) is then inert, V_{MM} and V_{ub} revert to their conventional (treat-W-as-known) values, and $\bar{Q}_M$ reverts to the fixed-weight trace. The estimated-weight law differs from current practice exactly through this channel. It enters CFI twice: through V_{uu} (the RMSEA-metric piece, where the channel is large because RMSEA's metric *is* the weight) and through the baseline objects $\hat{\delta}_b, V_{bb}$. One might expect the former to dominate and CFI to inherit RMSEA's large weight sensitivity, but empirically the *net* effect on $\widehat{\mathrm{Var}}(\widehat{\mathrm{CFI}})$ is small: the Monte-Carlo check puts the γ share at only $\pm$3–8% (versus -12% to -80% for RMSEA). The ratio structure and the $\hat{\delta}_b^2$ scaling attenuate the channel, and the $\hat{\delta}_b$ shift under a fixed weight largely cancels it, so CFI sits at the CRMR end of the spectrum: its estimated- and fixed-weight intervals nearly coincide.

7 Boundary and regularity

- **User boundary.** As $\hat{\delta}_u \downarrow 0$ (correct or near-correct user model) $\widehat{\mathrm{CFI}} \to 1$ sits on the parameter-space boundary, the truncation $\max(0,\cdot)$ activates, and the normal approximation degrades. The interval is clipped at 1 and flagged. This is the RMSEA close-fit boundary made sharper by CFI's hard ceiling.

- **Baseline boundary.** If $\hat{\delta}_b \le 0$ (independence model fits; essentially uncorrelated data) the ratio is undefined: report $\widehat{\mathrm{CFI}} = 1$, $\widehat{\mathrm{TLI}}$ and both intervals `NaN`, with a warning. The delta method requires $\hat{\delta}_b$ bounded away from zero, which holds whenever the baseline is a genuine null.

- **Shared metric.** $g_u'\Gamma_x g_b$ uses the *same* Γ_x for both gradients; the implementation must assert the two profile fits returned a common Γ_x (they do, being data-only) before forming the cross-term.

- **TLI variance is ill-conditioned at strong misfit.** Corollary 2 treats $c = \bar{Q}_b/\bar{Q}_u$ as fixed, but $\bar{Q}_u$ is a signed trace that can be small (and near its sign-change cancellation) when the user model misfits strongly, giving c^2 a heavy right tail. The Monte-Carlo check finds the TLI interval calibrated at weak/moderate misfit but *over-covering* at strong misfit (analytic variance $\sim 8\times$ the empirical), i.e. conservative, not anti-conservative. The TLI *point* stays unbiased. CFI carries no such division and is the index to trust for an interval; a stabilized c for TLI is future work.

- **Groups.** The pooling is in Section 8.

8 Multi-group

For G groups with sizes n_b, $N = \sum_b n_b$, the first-stage vectors $\hat{x}_b$ are independent across groups, $\sqrt{n_b}(\hat{x}_b - x_{0,b}) \xrightarrow{d} \mathcal{N}(0, \Gamma_{x,b})$. The fit minimizes the pooled discrepancy $F = \sum_b (n_b/N) F_b$, and every statistic in this note is a sample-size-weighted sum of per-group criteria,

$$T = N \sum_b \tfrac{n_b}{N}\,\mathrm{crit}_b = \sum_b n_b\,\mathrm{crit}_b \qquad (\mathrm{crit}_b = F_b \text{ for RMSEA, } G_b \text{ for CRMR}). \tag{10}$$

Each crit_b depends on $\hat{x}_b$ only, with per-group envelope gradient g_b. Because the groups are independent, the joint law collapses to a *block-diagonal* sum:

$$\mathrm{Var}(T) = \sum_b n_b\, g_b'\,\Gamma_{x,b}\, g_b, \qquad \mathrm{Cov}(T^{(1)}, T^{(2)}) = \sum_b n_b\, g_b^{(1)\prime}\,\Gamma_{x,b}\, g_b^{(2)}, \tag{11}$$

the second line being the CFI/TLI cross-term with the user and baseline gradients (cross-group terms vanish). The profile machinery already returns the block-diagonal $\Gamma_x = \mathrm{blkdiag}_b\,\Gamma_{x,b}$ and the pooled signed trace $\bar{Q} = \mathrm{tr}(Q\Gamma_x)$ (the two-metric core scales each block's Jacobian by $\sqrt{n_b/N}$, giving the correct pooled trace), so $\hat{\delta} = T - \bar{Q}$ is unchanged. The baseline is the per-group independence model: $T_b = \sum_b n_b F_{b,\mathrm{base}}$, $df_b = \sum_b \mathrm{ncorr}_b$, built as G analytic blocks through the same primitive with the *shared* per-group $\Gamma_{x,b}$.

In code (11) is one quadratic form: stack the per-group gradients into $g_\star$ with block b equal to $\sqrt{n_b/N}\, g_b$, so $g_\star'\Gamma_x g_\star = \frac{1}{N}\,\mathrm{Var}(T)$ matches the single-group $g'\Gamma_x g$ exactly (with $n_1 = N$ it is the same expression). The fixed-weight comparator zeros the γ coordinates *within each block* (global indices $[\,\mathrm{off}_b + m_b,\ \mathrm{off}_b + 2m_b)$), not a single trailing γ band. CFI, TLI, the ratio delta-method, and the boundary handling are otherwise identical with the pooled $\hat{\delta}$'s and (11).

9 Implementation map

1. User profile via `ordinal_dwls_profile_rmsea` $\Rightarrow Q_u, \Gamma_x, T_u, \bar{Q}_u$; form g_u from (4) with the fitted residual d_u (as the RMSEA path already does).

2. Baseline profile from the analytic independence block: jacobian $D_b = [\,I_{nth}; 0\,]$, residual d_b, weight γ, shared Γ_x, Gauss–Newton bread $D_b'WD_b$, $K_b = I$. Route through `weighted_moment_profile_rmsea_estimated_weight` $\Rightarrow Q_b, T_b, \bar{Q}_b$; cross-check T_b against `ordinal_baseline_chi2`. Form g_b from (4) with d_b.

3. $\hat{\delta}_u = T_u - \bar{Q}_u$, $\hat{\delta}_b = T_b - \bar{Q}_b$; $V_{uu} = Ng_u'\Gamma_x g_u$, $V_{bb} = Ng_b'\Gamma_x g_b$, $V_{ub} = Ng_u'\Gamma_x g_b$.

4. CFI (5), (6); TLI (7), (8); intervals per Corollaries 1–2 with the boundary handling of Section 7.

5. `estimated_weight=false` zeros the γ blocks of Γ_x before steps 3–4 (fixed-weight comparator).